

/ Tests pytest

Optimized testing with different operating systems and python versions

```
- name: Login into the GitHub container registry
  uses: docker/login-action@f4ef78c080cd8ba55a85445d5b36e214a81df20a
  with:
    password: ${{ secrets.GITHUB_TOKEN }}
    registry: ghcr.io
    username: ${{ secrets.GH_USERNAME }}
- env:
  LICENSE_SERVER: ${{ secrets.LICENSE_SERVER }}
  name: Pull, launch and verify desired Ansys service
  run: '...'
- name: Run pytest with desired markers and extra arguments
  uses: ansys/actions/tests-pytest@main
  with:
    pytest-extra-args: ${{ matrix.cfg.extra-args }}
    pytest-markers: ${{ matrix.cfg.markers }}
    requires-xvfb: true
```

Ref: [Optimized testing with different operating systems and Python versions](#)

Testing library with different operating systems and python versions

```
- name: Run pytest
  uses: ansys/actions/tests-pytest@main
  with:
    pytest-extra-args: --cov=ansys.<library>
    --cov-report=term --cov-report=html:.cov/html
    pytest-markers: -k 'mocked'
```

Ref: [Testing library with different operating systems and Python versions](#)

/ Hk migrate fork pr

Migrate fork pr to main repository

```
- name: Migrate fork PR
  uses: ansys/actions/hk-migrate-fork-pr@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-username: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    comment-id: ${{ github.event.comment.id }}
    github-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
    pr-number: ${{ github.event.issue.number }}
    team-slug: pyansys-maintainers
    user-triggering: ${{ github.event.comment.user.login }}
```

Ref: [Migrate fork PR to main repository](#)

/ Check actions security

Check actions security

```
- uses: ansys/actions/check-actions-security@main
  with:
    auditing-level: high
    generate-summary: true
    token: ${{ secrets.GITHUB_TOKEN }}
    trust-ansys-actions: true
```

Ref: [Check actions security](#)

/ Doc build

Installing additional system dependencies for building documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
  with:
    dependencies: graphviz mermaid-cli
```

Ref: [Installing additional system dependencies for building documentation](#)

Building library documentation that using xvfb

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
  with:
    requires-xvfb: true
```

Ref: [Building library documentation that using XVFB](#)

Building library documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
```

Ref: [Building library documentation](#)

/ Doc deploy pr

Deploy pr documentation

```
- uses: ansys/actions/doc-deploy-pr@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: ${{ env.DOCUMENTATION_CNAME }}
    maximum-pr-doc-deployments: 3
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy PR documentation](#)

/ Hk tag repository version

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@main
  with:
    gh-token: ${{ secrets.GITHUB_TOKEN }}
    release-version: ${{ github.event.release.tag_name }}
```

Ref: [Tag repository version](#)

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- env:
  SEMANTIC_VERSION: ${{ github.event.release.tag_name }}
  id: major_minor_version
  name: vX.Y version from tag
  run: 'MAJOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d
    'v' -f 2 | cut -d '.' -f 1)
    MINOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d
    'v' -f 2 | cut -d '.' -f 2)
```

```
echo
"MAJOR_MINOR_VERSION=v${MAJOR_VERSION}.${MINOR_VERSION}"
>> $GITHUB_OUTPUT
```

```
shell: bash
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@main
  with:
    gh-token: ${{ secrets.GITHUB_TOKEN }}
    latest-branch-name: release/${{
      steps.major_minor_version.outputs.MAJOR_MINOR_VERSION
    }}
```

Ref: [Tag repository version](#)

/ Check licenses

Check library dependencies ship with valid licenses

```
- name: Validate third party licenses
  uses: ansys/actions/check-licenses@main
  with:
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
```

Ref: [Check library dependencies ship with valid licenses](#)

/ Release pypi private

Release to private pypi

```
- name: Release to the private PyPI repository
  uses: ansys/actions/release-pypi-private@main
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${{ secrets.PYANSYS_PYPI_PRIVATE_PAT }}
    twine-username: __token__
```

Ref: [Release to private PyPI](#)

/ Check pr title

Check pull-request title follows conventional commits

```
- uses: ansys/actions/check-pr-title@main
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Check pull-request title follows conventional commits](#)

/ Doc changelog

Create changelog fragment

```
- uses: ansys/actions/doc-changelog@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Create changelog fragment](#)

/ Docker style

Docker style

```
- uses: ansys/actions/docker-style@main
  with:
    directory: docker
    error-level: 1
    recursive: true
```

Ref: [Docker style](#)

/ Doc deploy dev

Deploy developers documentation

```
- name: Deploy the latest documentation
  uses: ansys/actions/doc-deploy-dev@main
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy developers documentation](#)

/ Release pypi test

/ Build wheelhouse

Build wheelhouse for latest python versions

```
- name: Build a wheelhouse of the Python library
  uses: ansys/actions/build-wheelhouse@main
  with:
    library-name: <ansys-product-library>
    operating-system: ${ matrix.os }}
```

```
python-version: ${{ matrix.python-version }}
```

Ref: [Build wheelhouse for latest Python versions](#)

/ Hk package clean untagged

Perform untagged versions cleanup

```
- name: Perform untagged versions cleanup
  uses: ansys/actions/hk-package-clean-untagged@main
  with:
    package-name: mypackage
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Perform untagged versions cleanup](#)

/ Code style

Running code style checks

```
- name: Run PyAnsys code style checks
  uses: ansys/actions/code-style@main
```

Ref: [Running code style checks](#)

/ Doc style

Running documentation style checks

```
- name: Running documentation style checks
  uses: ansys/actions/doc-style@main
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Running documentation style checks](#)

/ Check vulnerabilities

Check library vulnerabilities

```
- uses: ansys/actions/check-vulnerabilities@main
  with:
    dev-mode: ${{ github.ref != 'refs/heads/main' }}
    python-package-name: ansys-<product>-<library>
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Check library vulnerabilities](#)

/ Hk automerge prs

Automerging dependabot and pre-commit.ci prs

```
- name: Automerging dependabot and pre-commit.ci PRs
  uses: ansys/actions/hk-automerge-prs@main
  with:
    approver: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    approver-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Automerging dependabot and pre-commit.ci PRs](#)

/ Release github

Release to github

```
- name: Release to GitHub
  uses: ansys/actions/release-github@main
  with:
    library-name: ansys-<product>-<library>
```

Ref: [Release to GitHub](#)

/ Doc deploy stable

Deploy stable documentation

```
- name: Deploy the stable documentation
  uses: ansys/actions/doc-deploy-stable@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy stable documentation](#)

/ Build ci wheels

Build a c-extension library

```
- name: Build a C-extension library wheel artifacts
  uses: ansys/actions/build-ci-library@main
```

Ref: [Build a C-extension library](#)

/ Doc deploy custom path

Deploy documentation to a custom path

```
- name: Deploy the documentation to a custom subdirectory
  uses: ansys/actions/doc-deploy-custom-path@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    custom-path: nightly
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy documentation to a custom path](#)

/ Build library

Build library basic example

```
- name: Build library source and wheel artifacts
  uses: ansys/actions/build-library@main
  with:
    library-name: ansys-<product>-<library>
```

Ref: [Build library basic example](#)

/ Release pypi public

/ Hk package clean except

Perform versions cleanup - except certain tags

```
- name: Perform versions cleanup - except certain tags
  uses: ansys/actions/hk-package-clean-except@main
  with:
    package-name: mypackage
    tags-kept: latest, latest-unstable
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Perform versions cleanup - except certain tags](#)

/ Doc deploy changelog

Update changelog for new tag

```
- uses: ansys/actions/doc-deploy-changelog@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Update CHANGELOG for new tag](#)